

Exerciti

① Pentru aplicațiile liniare următoare determinati
nucleul, imaginea (bază + dim.) și matricea asociată
în bazele canonică

a) $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y \\ x-y \\ x+2y \end{pmatrix}$

$$\text{Ker } f = \{v \in \mathbb{R}^3 \mid f(v) = 0_{\mathbb{R}^3}\}$$

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow f(v) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x+y=0 \\ x-y=0 \\ x+2y=0 \end{cases} \Rightarrow \begin{matrix} x=y=0 \\ z=\alpha, \alpha \in \mathbb{R} \end{matrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \alpha, \alpha \in \mathbb{R} \Rightarrow \text{Ker}(f) = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \alpha \mid \alpha \in \mathbb{R} \right\}$$

$$\text{Ker}(f) = \text{Sp} \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \text{dim Ker}(f) = 1$$

$$\text{Im}(f) = \{w \in \mathbb{R}^3 \mid \exists v \in \mathbb{R}^3 \text{ a.t. } f(v) = w\}$$

$$\dim_{\mathbb{R}} \text{Ker}(f) + \dim_{\mathbb{R}} \text{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}^3 \quad (\text{dimension})$$

$$\dim_{\mathbb{R}} \operatorname{Im}(f) = 2$$

$$y_{mf} = S_P \left\{ f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right); f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right); f\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) \right\} = S_P \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}; \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$b_{\text{inf}} = \text{Sp} \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$u(f) = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}$$

$$f\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = u_{11}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + u_{21}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + u_{31}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} u_{11} \\ u_{21} \\ u_{31} \end{pmatrix}$$

$$f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = u_{12}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + u_{22}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + u_{32}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} u_{12} \\ u_{22} \\ u_{32} \end{pmatrix}$$

$$f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = u_{13}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + u_{23}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + u_{33}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} u_{13} \\ u_{23} \\ u_{33} \end{pmatrix}$$

! $u(f)$ nu garanteaza in coeficientii sistemului folosit la gasirea nucleului ($\ker(f)$)

b) $f: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ $f\left(\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}\right) = \begin{pmatrix} x+2y+2z-w \\ 3x+6y+5z \\ x+2y+z+2w \end{pmatrix}$

$$\ker(f) = \left\{ v \in \mathbb{R}^3 \mid f\left(\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}\right) = 0 \right\}$$

$$\begin{cases} x+2y+2z-w=0 \\ 3x+6y+5z=0 \\ x+2y+z+2w=0 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{pmatrix} \xrightarrow[L_3-L_1]{L_2-3L_1} \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{pmatrix} \Rightarrow \begin{cases} x+2y+2z-w=0 \\ -z+3w=0 \end{cases}$$

$$\Rightarrow \begin{cases} x+2y+5w=0 \\ z=3w \end{cases}$$

Notăm $y = \alpha$ și $w = \beta$ nec. secundare $\Rightarrow \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -2\alpha-5\beta \\ \alpha \\ 3\beta \\ \beta \end{pmatrix}$
 $\alpha, \beta \in \mathbb{R}$

$$\ker(f) = \left\{ \begin{pmatrix} -2\alpha-5\beta \\ \alpha \\ 3\beta \\ \beta \end{pmatrix} \mid \alpha, \beta \in \mathbb{R} \right\} \Rightarrow \ker(f) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -5 \\ 0 \\ 3 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \dim_{\mathbb{R}} \ker(f) = 2$$

$$\dim_{\mathbb{R}} \ker(f) + \dim_{\mathbb{R}} \operatorname{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}^4 \quad \Bigg| \quad \Rightarrow \dim_{\mathbb{R}} \operatorname{Im}(f) = 2$$

$$\begin{aligned} \operatorname{Im}(f) &= \operatorname{Sp} \left\{ f \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}; f \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}; f \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}; f \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\} = \\ &= \operatorname{Sp} \left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}; \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix}; \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}; \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right\} = \operatorname{Sp} \left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}; \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right\} \end{aligned}$$

$$\mathcal{M}(f) = \begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{pmatrix}$$

c) $f: \mathbb{R}^4 \rightarrow \mathcal{M}_2(\mathbb{R}) \quad f \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} x-y & x-z \\ x+z+w & x+y+w \end{pmatrix}$

$$\begin{cases} x-y=0 \Rightarrow x=y \\ x-z=0 \Rightarrow z=x \\ x+z+w=0 \\ x+y+w=0 \end{cases} \quad \Bigg| \quad \begin{matrix} \\ \\ \Rightarrow 2x+w=0 \end{matrix} \quad \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} \alpha \\ \alpha \\ \alpha \\ -2\alpha \end{pmatrix} \quad \alpha \in \mathbb{R}$$

für $x=\alpha$ nec. secund.

$$\ker(f) = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ -2 \end{pmatrix} \right\} \Rightarrow \dim_{\mathbb{R}} \ker(f) = 1$$

$$\dim_{\mathbb{R}} \operatorname{Im}(f) = 4 - 1 = 3$$

$$\operatorname{Im}(f) = \operatorname{Sp} \left\{ \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}; \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}; \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{L_2-L_1 \\ L_3-L_1 \\ L_4-L_1}} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{L_3-L_2 \\ L_4-2L_2}} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{pmatrix}$$

vec. lin. indep (primäres und sekundäres)

$$\text{Im} f = \text{Sp} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}; \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}; \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$M(f) = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

② Fie $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ aplicatie liniara

$$f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y-z \\ x-y-z \\ x+y+z \end{pmatrix} \quad \text{verificati daca } f \text{ inj, surj, bi}$$

$$\text{Ker}(f) = \left\{ v \in \mathbb{R}^3 \mid f(v) = 0 \right\}$$

$$f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{cases} x+y-z=0 \\ x-y-z=0 \\ x+y+z=0 \end{cases} \xrightarrow{y=z} \begin{cases} x+z-z=0 \\ x-y-z=0 \\ x+y+z=0 \end{cases} \xrightarrow{y=-z} \begin{cases} x=0 \\ x-y-z=0 \\ x+y+z=0 \end{cases} \xrightarrow{2x=0 \Rightarrow x=0} \begin{cases} y=z \\ y=-z \\ x=0 \end{cases} \Rightarrow y=z=0$$

$$\Rightarrow \text{Ker}(f) = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\} \Rightarrow f \text{ injectiv}$$

$$\text{Im}(f) = \text{Sp} \left\{ f \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}; f \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; f \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} = \text{Sp} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}; \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}; \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\dim_{\mathbb{R}} \text{Ker}(f) + \dim_{\mathbb{R}} \text{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}^3 \Rightarrow$$

$$0 + \dim_{\mathbb{R}} \text{Im}(f) = 3 \Rightarrow \dim_{\mathbb{R}} \text{Im}(f) = 3 \quad \left. \begin{array}{l} \dim_{\mathbb{R}} \mathbb{R}^3 = 3 \end{array} \right\} \Rightarrow$$

$$\dim_{\mathbb{R}} \text{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}^3 \Rightarrow f \text{ surj.}$$

$$f \text{ inj } \wedge f \text{ surj} \Rightarrow f \text{ bi.}$$

③ Fie aplicația liniară $f: \mathbb{R}^3 \rightarrow M_2(\mathbb{R})$ cu proprietatea
 că $f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$; $f\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}$ și $\text{Ker}(f) = \left\{ \begin{pmatrix} a \\ -a \\ a \end{pmatrix} \mid a \in \mathbb{R} \right\}$
 Determinați $\text{Im}(f)$ (dim + bază) și $M(f)$

$$\dim \text{Ker}(f) = 1 \Rightarrow \dim \text{Im}(f) = 3 - 1 = 2 \Rightarrow$$

$$\Rightarrow \text{Im}(f) = \text{Sp} \left\{ \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}; \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \right\} \text{ lin. indep. } \rightarrow \text{ e bază}$$

Varianta 1

$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = ?$ Verificare dacă vectorii $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}; \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$
 formează bază în $\mathbb{R}^3$

$$\begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{bază} \Rightarrow$$

$\Rightarrow \forall \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$ avem $\alpha_1, \alpha_2, \alpha_3$ a.f.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} \alpha_1 + 2\alpha_2 + \alpha_3 = x \\ \alpha_2 - \alpha_3 = y \\ \alpha_3 = z \end{cases} \Rightarrow \begin{cases} \alpha_1 = x - 2y - 3z \\ \alpha_2 = y + z \end{cases}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x - 2y - 3z) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (y + z) \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \quad \nearrow$$

f aplicație liniară

$$\Rightarrow f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x - 2y - 3z) f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (y + z) f\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z f\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x - 2y - 3z) \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} + (y + z) \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} + z \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y & x-z \\ -x+3y+4z & -y-z \end{pmatrix} \quad M(f) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 3 & 4 \\ 0 & -1 & -1 \end{pmatrix}$$

Varianța 2

$$f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$$

$$f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = ?$$

$$f\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}$$

$$f\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = 2f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix}$$

$$f\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$f\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = ?$$

$$f\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + f\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow f\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 4 & -1 \end{pmatrix}$$

$$f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x f\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y f\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + z f\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

④ fie $f: \mathbb{R}_{\leq 2}[x] \rightarrow \mathbb{R}_{\leq 3}[x]$ o aplicație liniară cu prop. că

$$f(1) = 1; f(x) = x^2 + 1, f(x^2 + 1) = x^3$$

a) Să se determine $f(P)$, $P = a + bx + cx^2$, $a, b, c \in \mathbb{R}$

$$f(1+x+x^2) = f(x^2+1) + f(x) = x^3 + x^2 + 1$$

$$f(a+bx+cx^2) = f[a+bx+c(x^2+1)-c] = (a-c)f(1) + cf(x^2+1) + bf(x) = a-c + cx^3 + bx^2 + b = a+b-c + bx^2 + cx^3$$

b) Verif, $\text{Im} f = ?$

$$\text{Ker}(f) = \{P \in \mathbb{R}_{\leq 2}[x] \mid f(P) = 0\}$$

$$f(P(x)) = 0 \quad \forall x \in \mathbb{R} \Rightarrow \begin{cases} c=0 \\ b=0 \\ a+b-c=0 \rightarrow a=0 \end{cases} \Rightarrow \text{Ker}(f) = \{0\}$$

$$\Rightarrow \dim_{\mathbb{R}} \ker(f) = 0$$

$$\dim_{\mathbb{R}} \ker(f) + \dim_{\mathbb{R}} \operatorname{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}_{\leq 2}[x] \quad \Big| \Rightarrow$$

$$\Rightarrow \dim_{\mathbb{R}} \operatorname{Im}(f) = 3$$

$$\operatorname{Im}(f) = \{f(1); f(x); f(x^2-1)\} \quad \Big| \Rightarrow \operatorname{Im}(f) = \operatorname{Sp}_{\mathbb{R}} \{1; x^2+1; x^3-1\}$$

baza

$$f(x^2-1) = x^3-1$$

$$5) f: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x+3y-z \\ y+z \\ x+2z \end{pmatrix}$$

$$a) \mathcal{M}(f)$$

$$\mathcal{M}(f) = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix}$$

$$b) P_{B_C}^B \quad B = \left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{v_1}; \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_2}; \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v_3} \right\} \quad \mathcal{M}_B(f) = ?$$

$$P_{B_C}^B = \left([v_1]_{B_C} \quad [v_2]_{B_C} \quad [v_3]_{B_C} \right) \quad [v_1]_{B_C} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$$

$$v_1 = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3$$

$$(e_1 \quad e_2 \quad e_3; \quad v_1 \quad v_2 \quad v_3)$$

$$\begin{pmatrix} 1 & 0 & 0 & : & 1 & 1 & 1 \\ 0 & 1 & 0 & : & 0 & 1 & 1 \\ 0 & 0 & 1 & : & 0 & 0 & 1 \end{pmatrix}$$

$P_{B_C}^B$

$$\mathcal{M}_B(f) = P_B^{B_C} \cdot \mathcal{M}(f) \cdot P_{B_C}^B$$

$(P_{B_C}^B)^{-1}$